Applied Computer Systems
ISSN 2255-8691 (online)
ISSN 2255-8683 (print)
December 2024, vol. 29, no. 2, pp. 22–29
https://doi.org/10.2478/acss-2024-0018
https://content.sciendo.com

Age Prediction from Facial Images Using Deep Learning Architecture

Hai Thanh Nguyen[1*], Linh Thuy Thi Pham[2], Dung Thi Dang[3], Son Nguyen Huynh[4], Phu Hao Dang[5], Quoc Thien Huynh Nguyen[6]
[1]*College of Information and Communication Technology, Can Tho University, Can Tho, Vietnam*
[2–6]*Can Tho University of Technology, Can Tho, Vietnam*

Abstract **– Predicting age and gender through images is a common computer vision problem with many practical applications. However, this problem faces many difficulties because a person's age can be affected by genetics, living environment, diet, health, gender, and other factors. Therefore, the accuracy of the prediction model may decrease due to the enormous diversity and variability in the data. In this study, we use three models, including Unet, MobileNets, and EfficientNets, to test the performance of predicting a person's age and gender through their photos. In addition, we also adjust the learning rate parameter to find optimal performance. The best results for gender prediction are achieved by the Unet model with the highest accuracy of 97.22 %, and the MobileNets model gives age prediction results with MAE = 2.248, learning rate 0.001 for optimal performance in the models of our study.**

Keywords **– Age prediction, computer vision, face image, Unet.**

I. Introduction

The human face expresses a diversity of shapes, including features such as the eyes, nose, mouth, chin, eyebrows, and skin colour [1]. When they are combined with facial expressions and other contours, these characteristics create a unique human face, enabling us to recognise and distinguish individuals. This distinctiveness is evident across all languages worldwide, influencing how we interact with someone young versus elderly. Recognising a person's age and gender from the face can help us communicate more appropriately, rather than relying solely on estimates from external facial features. Additionally, this has critical applications in many different fields. In security and law enforcement [2], determining age and gender through facial images can aid in criminal identification and detecting falsified personal information, thereby addressing public security concerns. In the medical field [3], information about a person's age and gender can be crucial for diagnosing illnesses, treatment, and health management, especially in early aging detection. In technology, this can be used to personalise user experiences on online platforms, from marketing to fraud detection [4].

Estimating a person's age and gender based on their face remains challenging due to differences in facial features and appearance, and the lack of comprehensive databases. Recently, convolutional neural network (CNN) methods have been applied for age estimation and image classification with notable improvements [5], [6]. In forensic research [7], to predict gender and age groups from human eye area images, VGG16 was used to classify age groups. The results showed that the VGG16 model effectively predicted gender and age from eye area images.

This study used the UTKFace dataset, which contains over 20 000 face images. We aim to build a model to classify gender and age from faces. We implemented three models, including Unet, MobileNets, and EfficientNets. We tested and compared different learning rates, including 0.001, 0.005, and 0.01.

The remaining parts of the article are organised as follows: Section II presents research on age and gender prediction. Section III describes in detail the dataset, data pre-processing, and algorithms used for comparison. Results and explanations of each algorithm are presented in Section IV. Section V summarises potential future research directions and concludes with the research implications.

II. Related Work

There have been many studies on prediction, including predicting age and gender based on their faces. Some studies include [8] the Consistent Rank Logits (CORAL) method to adapt popular CNN architectures and compare the results using data warehouses, such as MORPH-2, CACD, and AFAD. The MORPH-2 data set contains 55 608 face images ranging from 16–70 years old, CACD data contains 159 449 images ranging from 14–62 years old, and Asian face data (AFAD) contains 165 501 faces between 15–40 years old. They chose the ResNet-34 architecture and the results were: OR-CNN method (AVG ± SD of MAE: 2.83 ± 0.03, AVG ± SD of RMSE: 3.97 ± 0.11), CORAL-CNN method (AVG ± SD of MAE: 2.64 ± 0.02, AVG ± SD of RMSE: 3.65 ± 0.04). The experimental results of the study showed that CORAL improved the performance expectations of the convolutional neural network for age estimation on three independent datasets.

* Corresponding author's e-mail: nthai.cit@ctu.edu.vn
Article received 2024-07-14; accepted 2024-10-21

The study [9] compared the effect of initialising the AdienceNet, CaffeNet, Googlenet, and VGG-16 models with pre-trained weights on two real-world datasets, including the Adience database that contains 26 580 images on 2284 subjects with binary gender labels and one label from eight different age ranges. They compared image pre-processing, model bootstrapping, and architectural choices on the Adience dataset and discussed how they affected performance. Using LRP to visualise how the model interacts with specific input samples directly, they demonstrated that bootstrapping the model appropriately through back-mining the overfitting phenomenon leads to a view of comprehensive input. With the combination of simple pre-processing steps, the study achieved superior performance for gender classification on the Adience data test set.

In the study [10], the authors used YOLO to locate faces in images with a Web application written in the Python programming language. They used a model based on the EfficientNet architecture to predict age. The MIVIA Age Dataset has 575 073 photos of more than 9000 people of different ages. Face recognition accuracy is 94.6 %; age recognition MAE index is 2.89. A deep learning solution was introduced to estimate age from a facial image without using facial landmarks [11]. They crawled 523 051 facial images from IMDb and Wikipedia websites to create IMDB-WIKI. The authors used the Deep EXpectation (DEX) system to estimate age. First, they selected an off-the-shelf face detector to obtain the face's position and size (scale) in each image. Next, they found that performance increased when considering the context around the face, and they expanded face detection by taking 40 % more width and height of the face at all edges. Then, they used a Convolutional Neural Network of the VGG-16 architecture pre-trained on ImageNet to classify the images. Finally, they estimated the age by screening the Softmax expectation value. When identifying age, the MAE index was 3.306.

A scheme for extracting aging characteristics and automatic age estimation is presented in [12] with the basic idea of learning by embedding a low-dimensional aging manifold using the appropriate subspace of the learning method. They then designed a new method, locally adjusted regression (LARR), to learn and predict aging patterns. They tested the FG-NET database for age estimation and age-specific face recognition with 1002 high-resolution colour or grayscale face images with significant variations in lighting, poses, and expressions of 82 people with ages ranging from 0 to 69 and an age gap of up to 45 years. When identifying age, MAE index was 5.04.

In [13], the authors focus on using the transfer learning method to classify the dural age group; 2000 Sclera images are collected from 250 people of different ages, and the Otsu threshold is used to segment images using morphological processes. The segmented images were trained and tested on four different pre-trained models (VGG16, ResNet50, MobileNetV2, EfficiencyNet-B1), which were compared based on different performance metrics in which ResNet-50 was shown to outperform the others, resulting in an accuracy, precision, recall and F1-score of 95% while VGG-16, EffcientNetB1, and MobileNetV2 had 94 %, 93 %, and 91 %, respectively. The study findings suggest an aging pattern in the dura mater that can be used to classify age.

A new framework for extracting age characteristics based on CNN, dimensionality reduction, and classification (SVR, PLS, CCA) was built [14] for estimating age. Compared to previous models based on CNN, this approach utilizes feature maps obtained in different layers to estimate activity instead of features obtained in the top layer. Experiments were conducted on two datasets, MORPH and FG-NET. When identifying age on MORPH and FG-NET, the MAE index was 4.77 and 4.26, respectively. The study [15] also proposed a multi-output CNN to solve the age and gender classification problem. High performance was achieved in both the MORPH dataset and the Asian Face Age dataset (AFAD). When identifying age on MORPH and AFAD, the MAE index was 3.27 and 3.34, respectively. Additionally, the authors proposed solving the ordinal regression problem using an end-to-end deep learning method. Specifically, an ordinal regression problem was transformed into a series of binary classification sub-problems collectively solved by the proposed multi-output CNN learning algorithm.

A nutritional recommendation system for users based on age and gender prediction was built [16] with the method of pre-processing data images before going through the feature extraction stage using a deep neural network (DNN). The system was evaluated by classification rate, precision, and recall using the Adience and UTKface datasets. On the other hand, the study [17] focused on age estimation, age classification, and gender classification from static images of an individual with 20 000 photos of the UTKface dataset. The results showed that simple linear regression trained on the extracted features outperformed CNN, ResNet50, and ResNeXt50 training for age estimation. Specifically, for the Separable Conv2D + Spatial dropout + Xavier uniform initialisation method in age estimation, MAE = 6.080; for age classification, the accuracy was 78.279 %; for gender classification, the accuracy was 91.269 %. Research [18] showed that improving the accuracy of models that determine age and sex from images was sometimes hampered by external factors such as lighting, makeup, and politics. Therefore, some models cannot accurately predict age for these reasons. In terms of methods, they performed the following tasks: Load data (IMDB-WIKI), Detect faces (using Viola-Jones algorithm), Crop and resize faces (227×227×3), Extract features typically using CNN, apply PCA to extracted features (PCA is applied to reduce the dimensionality of extracted features from 4096 to 500), train and test using CNN, visualise results and analyse. The results obtained were that for the gender prediction model, the accuracy was 95.5 %, MAE (0.871), MSE (0.8900), and for the age prediction model, the accuracy was 89.50 %, MAE (0.1441), MSE (0.3517).

The study [19] showed that gender prediction was possible through images recorded with the phone front camera. The authors used VGG and ResNet models as feature representations and pattern classifiers to predict gender on the large ImageNet dataset. The results showed that the VGG-16 model achieved 85.3 % accuracy using SVM on the eye region, including single eyes taken with iPhone 5s. The Classification Error Rate (MCR) and Misclassification Error Rate (FCR) were 91.7 % and 73.7 %, respectively. When combining SVM with MLP at the score level, accuracy increased slightly to 85.7%, while MCR and FCR decreased to 91.5 % and 77.6 %. There were no significant differences between the VGG-16 and VGG-19 models. In the ResNet Model, the best accuracy was about 87.3% achieved by SVM on the Oppo device. The classification error rate (MCR) and misclassification error rate (FCR) were 90.2 % and 84.8 %, respectively. Combining SVM with MLP at the point level increased the accuracy to 89.0 %. MCR and FCR were 87.5 % and 88.0 %, respectively. Studies have shown many positive results in identifying a person's age and gender through real-time images and videos. In this study, we performed age and gender prediction through three models: Unet, MobileNets, and EfficientNets. We tested the performance of the models through accuracy and the MAE index through epoch changes and learning rate. The implementation method is presented in detail in the next section.

III. Methods

This section introduces the research method, which consists of three parts. First, Section III-A presents the UTKFace dataset, which contains more than 8000 face images. Next, we introduce the image pre-processing steps since the initial data is imbalanced. After that, we train our data using three models: Unet, EfficientNet, and MobileNets. In Section III-C, we describe the architecture and settings of each model. Section III-D provides information about programming language settings, necessary libraries, and computer configuration. Finally, the evaluation parameters in the model include MAE for age prediction and accuracy for gender prediction. Figure 1 describes in detail the implementation architecture of our model.

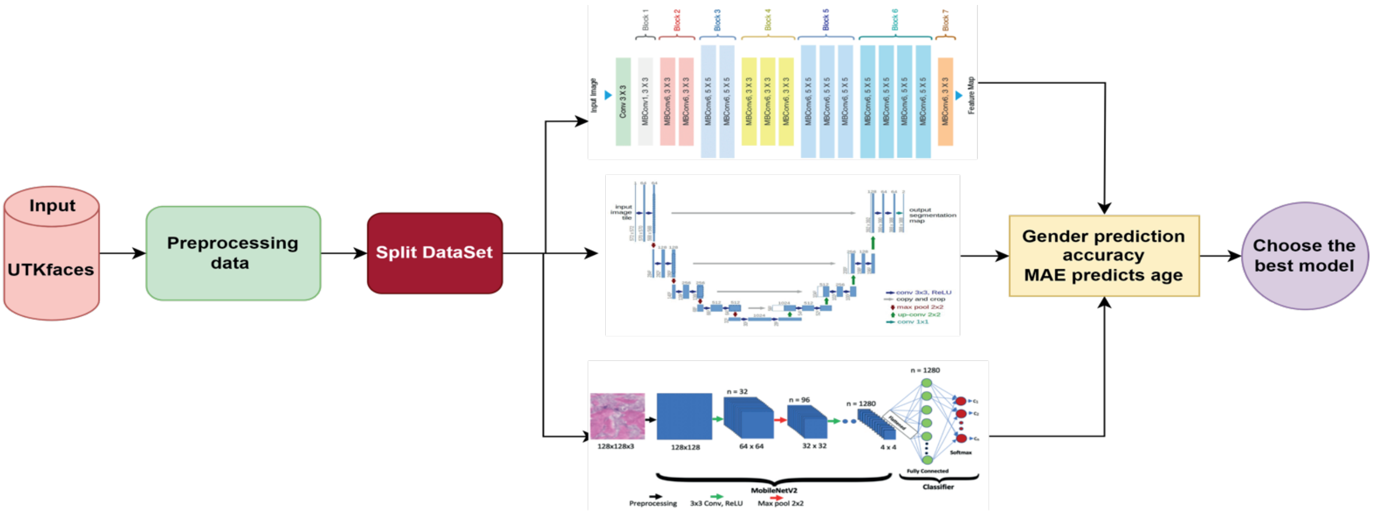


Fig. 1. Our proposed model architecture.

A. Dataset

We utilised the UTKFace dataset[2] comprising over 20 000 annotated facial images, including age, gender, and ethnicity information, ranging from 0 to 116 years old. The age group of 20–30 years old constitutes the majority. The elderly group represents a minority compared to other age groups; hence, in this study, we excluded images with ages ranging from 86 to 100 to avoid age group imbalance. The dataset predominantly comprises individuals with white ethnicity (approximately 5000 images, over 50 %), followed by a minority of black individuals (only around 3–5 %), with other ethnicities accounting for fluctuating proportions ranging from 10–15 %. The female gender outweighs the male gender. Due to memory constraints, we only utilised the first 8000 images. Figure 2 depicts a detailed description of our dataset. The label of each image in the dataset has been annotated according to the structure [age]-[gender]-[race]-[date&time], with the extension in jpg format.

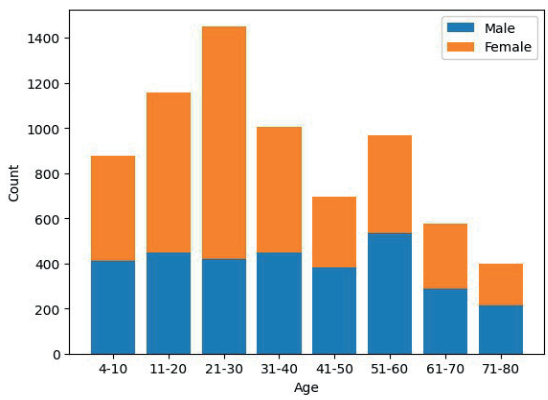


Fig. 2. Description of the dataset in our study.

In the dataset, [age] is an integer from 0 to 116 representing age, [gender] represents gender with 0 (male) or 1 (female), and [race] is an integer from 0 to 4 representing skin white, Black,

[2] https://susanqq.github.io/UTKFace/

Asian, Indian, [date&time] is in the format "yyyymmddHHMMSSSFFF", showing the date and time the image was collected in UTKFace.

B. Image Pre-processing

As mentioned earlier, the dataset of more than 9000 photos from the UTKFaces library is not uniform, and our goal is to compare the performance between three models: EfficientNet, U-Net, and MobileNet. We aim to determine which model is most suitable based on various factors such as parameters, performance, and accuracy. The dataset consists of photographs that have been cropped to show only the subject's face. Therefore, our data pre-processing focuses on rebalancing the dataset to ensure class uniformity.

Our image pre-processing steps include looping through all image files in the directory and extracting age and gender information from the image file names. After reading each image file, we use "ImageOps.fit" to adjust the image size to fit each model. Once the images are read and processed, we create a data frame from the collected data. We then remove unnecessary data, such as eliminating samples with ages below four years by selecting those with ages greater than four and excluding samples with ages outside the range of 4–80 or gender of 3. We randomly select a portion (30 %) of the samples under four years old and add them to the dataset to balance the data. Finally, we normalise the image data by dividing by 255 to ensure pixel values are in the range [0, 1]. These pre-processing steps help prepare the data for input into a machine-learning model to predict age and gender based on the images.

C. Classification and Object Detection

In this section, we apply machine learning methods to create a model that predicts a person's age and gender. The modelling process involves selecting from various models and employing different techniques during testing. In this case, we used models such as MobileNets [20], Unet [21], and EfficientNets [22] for prediction. We aim to identify the best classification model for this analysis problem, ensuring the highest possible accuracy and performance.

D. Environmental Settings

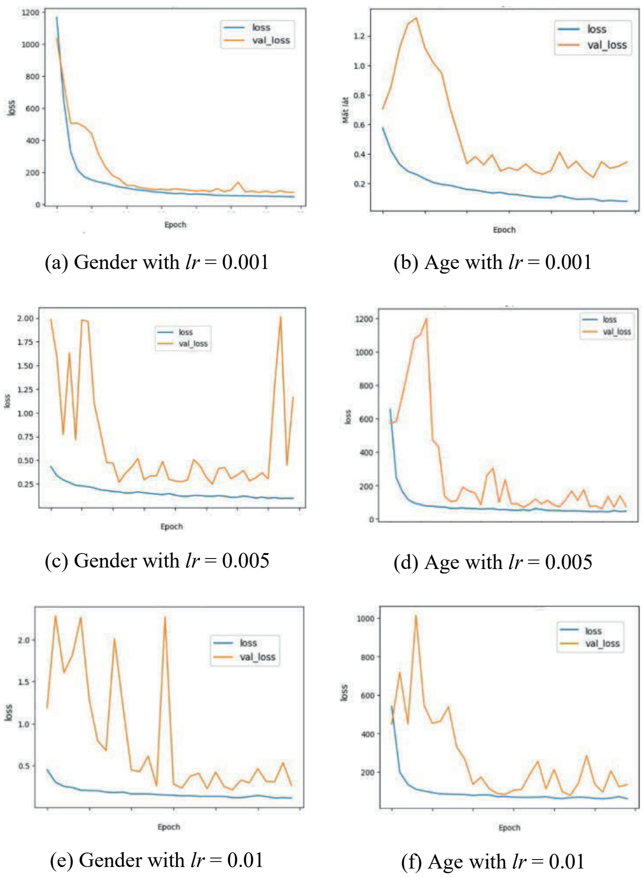
First, to run and train models with significant parameters, we train the model on Google Colab Pro, where the code is sent to Google's server, executed, and results are returned. We select the Tesla K80 GPU environment with approximately 12 GB of RAM and the V100 GPU. The Python programming language is used along with necessary libraries, including Keras in TensorFlow, Sklearn, Pandas, NumPy, Operating System, Python Imaging Library (PIL), TensorFlow, and segmentation models. The dataset is divided into four subsets to serve two goals: age identification and gender recognition. Each goal includes two datasets: a training dataset and a test dataset. For each test dataset, 20 % of representative data from each group is selected for testing. This study uses two parameters to evaluate the model's performance. For age prediction, we use the Mean Absolute Error (MAE). For gender prediction, we use accuracy. MAE is a measure commonly used in regression problems to assess the quality of a prediction model. It measures the average absolute difference between predicted and actual values – the lower the MAE value, the more accurate the model.

IV. Experimental Results

This section presents the results of the study's experimental scenarios. We set up the models with default parameters, including an input image size of 224×224, batch size of 64, 100 epochs, and a learning rate, which we varied to include 0.001, 0.005, and 0.01 to test the performance changes of the models. We present detailed results in Section IV-A for the Unet model, Section IV-B for the MobileNets model, and Section IV-C for the EfficientNets model. Additionally, we compare our study with previous studies in Section IV-D.

A. Performance of the Unet Model

To optimise the training process, we combined Unet with ResNet34, using it as a feature extraction encoder for the input images. By doing so, the model does not need to learn basic features from scratch, reducing the complexity of the training process and the total number of parameters that need to be trained. We experimented with different learning rates (lr), including 0.001, 0.005, and 0.01, to select the most suitable rate for the environment. A learning rate that is too high may cause the model to "jump over" the optimal point, while a learning rate that is too low may result in slow learning or cause the model to get stuck at a local optimum.



(a) Gender with $lr = 0.001$ (b) Age with $lr = 0.001$

(c) Gender with $lr = 0.005$ (d) Age with $lr = 0.005$

(e) Gender with $lr = 0.01$ (f) Age with $lr = 0.01$

Fig. 3. Loss function results when changing the learning rate in the Unet model.

The results in Fig. 3 demonstrate a discrepancy between the training and testing sets. This indicates that the model needs to generalise to new data as the learning rate increases. At a learning rate of 0.001, the model achieves lower errors for age (MAE = 4.722) and higher accuracy for gender (accuracy = 0.9722), compared to a learning rate of 0.01, where the errors increase to 6.3242 (MAE) for age, and gender accuracy decreases to 0.95978. This discrepancy suggests that higher learning rates hinder the model's training and testing dataset generalisation. Detailed results are shown in Table I. Figure 4 presents the prediction results using the Unet model.

TABLE I

PERFORMANCE OF THE UNET MODEL WHEN CHANGING THE LEARNING RATE

Learning Rate	Gender Acc.	Age MAE	Age RMSE
0.001	0.9722	4.722	5.8360
0.005	0.9602	6.578	7.6306
0.01	0.9598	6.324	7.5213

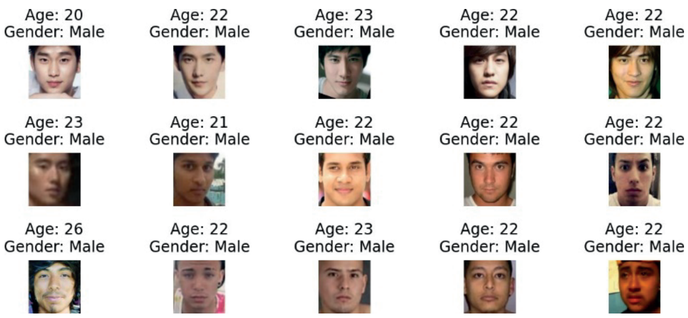


Fig. 4. Prediction results using the Unet model.

B. Performance of the MobileNets Model

With the MobileNets model, we explored different learning rates, including 0.001, 0.005, and 0.01. The results in Table II reveal that at a learning rate of 0.001, the best performance in predicting gender on both the training and validation datasets is achieved, with accuracies of 87.45 % and 87.86 %, respectively. The best MAE is 2.44824, and the RMSE is 3.4458. When using this model, all three balance charts exhibit relatively minor differences between the training and testing sets. At a learning rate of 0.005, the gender prediction accuracy drops to 56.43 %, with an MAE of 2.8214 and an RMSE of 3.856, lower than those at *lr* = 0.001.

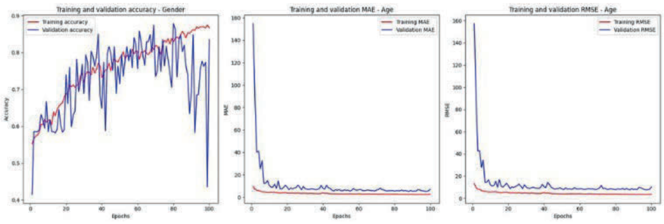
However, all three balance charts show an imbalance, with excellent results in the training set but inferior results in the testing set. At a learning rate of 0.01, the gender prediction accuracy further decreases to 56.18 %, with an MAE of 3.5468 and an RMSE of 4.6948.

All three balance charts exhibit more significant differences between the training and testing sets than those at *lr* = 0.001 and *lr* = 0.005, indicating relatively low accuracy in gender prediction. Figure 5 illustrates the loss function results for predicting age and gender at different learning rates. Figure 6 shows the prediction results using the MobileNets model.

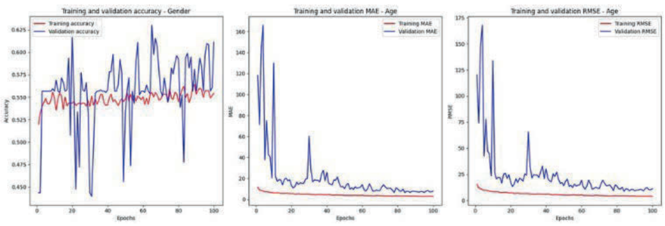
TABLE II

PERFORMANCE OF THE MOBILENETS MODEL WHEN CHANGING THE LEARNING RATE

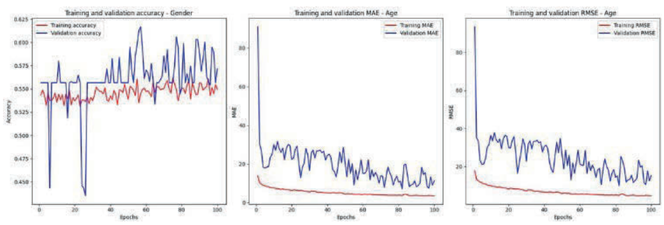
Learning Rate	Gender Acc.	Age MAE	Age RMSE
0.001	0.8745	2.4482	3.4458
0.005	0.5644	2.8214	3.8556
0.01	0.5618	3.5468	4.6948



(a) *lr* = 0.001



(b) *lr* = 0.005



(c) *lr* = 0.01

Fig. 5. Results of testing the accuracy in the MobileNet model with learning rate.

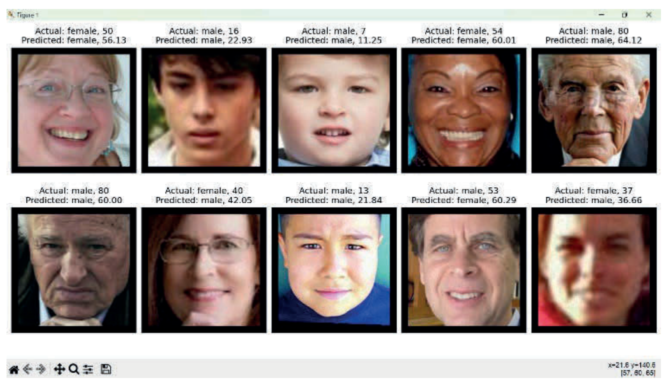


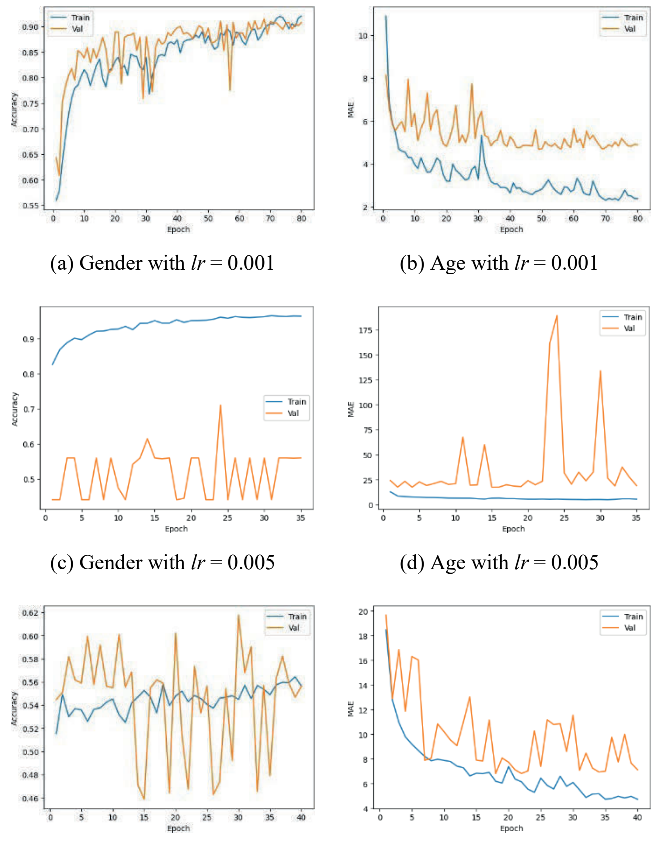
Fig. 6. Results before and after prediction using the MobileNet model.

C. Performance of the EfficientNets Model

Similarly to the previous two models, we experimented with different learning rates for the EfficientNets model. The results in Table III demonstrate that at a learning rate of 0.001, the best performance in predicting gender on both the training and validation datasets is achieved, with accuracies of 92.03 % and 91.47 %, respectively. Likewise, the best MAE scores are 2.3060 (train) and 4.6693 (val). Both balance charts show minor discrepancies between the training and testing sets when using this model. At a learning rate of 0.005, the best gender prediction performance on the training and validation datasets is achieved with 96.52 % and 70.99 % accuracy, respectively.

TABLE III
PERFORMANCE OF THE EFFICIENTNETB3 MODEL WHEN CHANGING THE LEARNING RATE

Learning Rate	Accuracy of Gender	Age MAE
0.001	0.9203	2.3060
0.005	0.9652	4.8394
0.01	0.5644	4.7328



(a) Gender with $lr = 0.001$ (b) Age with $lr = 0.001$

(c) Gender with $lr = 0.005$ (d) Age with $lr = 0.005$

(e) Gender with $lr = 0.01$ (f) Age with $lr = 0.01$

Fig. 7. Loss function results when changing the learning rate in the EfficientNetB3 model.

Similarly, the best MAE scores are 4.8394 (train) and 17.4116 (val). Both balance charts exhibit an imbalance, with excellent results in the training set but subpar results in the testing set. At a learning rate of 0.01, the best gender prediction performance on the training and validation datasets is achieved with 56.44 % and 60.20 % accuracy, respectively. Likewise, the best MAE scores are 4.7328 (train) and 6.8154 (val). Both balance charts show relatively minor discrepancies between the training and testing sets, but the accuracy in gender prediction is relatively low. Figure 7 illustrates the loss function results for predicting age and gender at different learning rates.

To test the model, we allowed it to recognise images randomly taken from the data warehouse featuring different ages and genders. Figure 8 shows the prediction results using the EfficientNet model. The results show that the subjects' genders are correctly identified; however, their ages differ by three years or more. This discrepancy may be attributed to the quality of the photos, but this difference is still within an acceptable range.

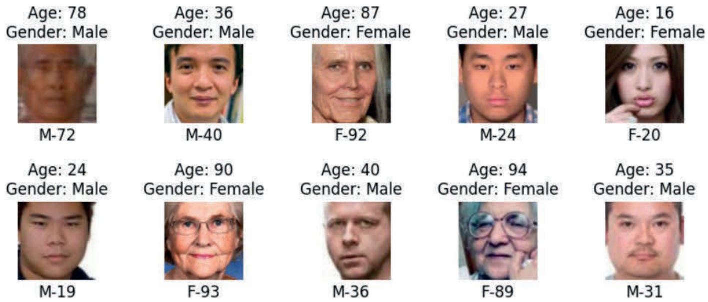


Fig. 8. Prediction results of the EfficientNet model.

D. Comparison with Previous Studies

Recently, convolutional neural network (CNN) methods have been applied for age estimation and image classification, and improvements have been noted. To demonstrate the performance of the models, we compare our study with previously conducted related studies. The studies we compared included the VGG-16 model [19], which achieved 85.3 % accuracy, and another study [7] that predicted gender with 95.73 % accuracy and age group with 99.41 % accuracy. Feature extraction using the CNN prediction model [18] achieved a gender accuracy of 95.5 %, with an MAE of 0.871 and an MSE of 0.8900. In contrast, the accuracy of the age prediction model is 89.50 %, with an MAE of 0.1441 and an MSE of 0.3517. The AutoML automatic deep learning model [23] predicts gender with an accuracy of 69.4 %. The Inception V3 model [24] has an accuracy of 91.82 %, and the YOLO model [10] has a face recognition accuracy of 94.6 %; in age recognition, the MAE index is 2.89. Finally, the deep convolutional neural network (DCNN) model [16] achieves an MAE of 6.080 for age estimation, an accuracy of 78.279 % for age classification, and an accuracy of 91.269 % for gender classification. In 2023, a study [7] built the VGG16 model that predicted gender with 95.73% accuracy and age group with 99.41 % accuracy. This study used the UTKFace dataset containing more than 20 000 face images. From the comparison results in Table IV, we see that our U-Net model gives the highest gender prediction result with 97.22 %, and the best MAE index is in the MobileNets model with 2.88214.

TABLE IV

COMPARISON OF THE MODEL PERFORMANCE WITH RELATED STUDIES

Model	**Accuracy**	**MAE**
Unet	97.220	4.722
MobileNets	87.450	2.821
EfficientNets	96.520	4.839
CNN ([18])	95.500	0.871
SVMs ([25])	79.700	-
VGG-16 ([19])	85.300	-
VGG16 [7]	95.730	-
AutoML ([23])	69.400	-
Inception V3 ([24])	91.820	-
Yolo ([10])	94.600	2.890
DCNN ([16])	91.269	6.080

V. CONCLUSION

Predicting age and gender has never been easy, and it faces many challenges, such as discrepancies in data, ambiguity in age and gender, and peripheral factors like lighting and photo background. The complexity and diversity of the data increase the difficulty of training the model for stable and optimal performance. In this study, we use models such as U-Net, MobileNets, and EfficientNets to address the problem of recognising and classifying age and gender based on images. The U-Net model achieves 97.22 % accuracy for gender prediction and an MAE of 4.722 for age prediction. With the MobileNets model, the accuracy is 87.45 % for gender prediction, with an MAE of 2.448 for age prediction. Finally, the EfficientNets model achieved 96.52 % accuracy for gender prediction and an MAE of 2.306 for age prediction. The best results were obtained when the learning rate was set to 0.001. In the future, we will continue to extract features based on each facial characteristic to support individual anti-aging efforts.

CONFLICT OF INTEREST STATEMENT

All authors declare that they have no conflicts of interest.

REFERENCES

[1] M. Maithri, U. Raghavendra, A. Gudigar, J. Samanth, P. D. Barua, M. Murugappan, Y. Chakole, and U. R. Acharya, "Automated emotion recognition: Current trends and future perspectives," *Computer Methods and Programs in Biomedicine*, vol. 215, p. 106646, 2022. https://doi.org/10.1016/j.cmpb.2022.106646

[2] T. L. Johnson, N. N. Johnson, V. Topalli, D. McCurdy, and A. Wallace, "Police facial recognition applications and violent crime control in U.S. cities," *Cities*, vol. 155, p. 105472, Dec. 2024. https://doi.org/10.1016/j.cities.2024.105472

[3] F. Mauvais-Jarvis, N. B. Merz, P. J. Barnes, R. D. Brinton, J.-J. Carrero,D. L. DeMeo, G. J. De Vries, C. N. Epperson, R. Govindan, S. L. Klein, et al., "Sex and gender: modifiers of health, disease, and medicine," *The Lancet*, vol. 396, no. 10250, pp. 565–582, 2020.

[4] A. Mutemi and F. Bacao, "E-commerce fraud detection based on machine learning techniques: Systematic literature review," *Big Data Mining and Analytics*, vol. 7, no. 2, pp. 419–444, 2024.

[5] A. Abdolrashidi, M. Minaei, E. Azimi, and S. Minaee, "Age and gender prediction from face images using attentional convolutional network," *arXiv:2010.03791*, Oct. 2020. https://doi.org/10.48550/arXiv.2010.03791

[6] D. Yi, Z. Lei, and S. Li, "Age estimation by multi-scale convolutional network," in *Asian Conference on Computer Vision*, 2014, pp. 144–158. https://doi.org/10.1007/978-3-319-16811-1_10

[7] Ö. F. Akmeşe, H. Çizmeci, S. Özdem, F. Özdemir, E. Deniz, R. Mazman, M. Erdoğan, and E. Erdoğan, "Prediction of gender and age period from periorbital region with VGG16," *Chaos Theory and Applications*, vol. 5, no. 2, pp. 105–110, July 2023. https://doi.org/10.51537/chaos.1257597

[8] W. Cao, V. Mirjalili, and S. Raschka, "Rank consistent ordinal regression for neural networks with application to age estimation," *Pattern Recognition Letters*, vol. 140, pp. 325–331, Dec. 2020.

[9] https://doi.org/10.1016/j.patrec.2020.11.008W. Samek, A. Binder, S. Lapuschkin, and K.-R. Muller, "Understanding and comparing deep neural networks for age and gender classification," in *2017 IEEE International Conference on Computer Vision Workshops (ICCVW)*, Venice, Italy, Oct. 2017, pp. 1629–1638. https://doi.org/10.1109/ICCVW.2017.191

[10] G. Castellano, B. D. Carolis, N. Marvulli, M. Sciancalepore, and G. Vessio, "Real-time age estimation from facial images using YOLO and EfficientNet," in *Computer Analysis of Images and Patterns*, N. Tsapatsoulis, A. Panayides, T. Theocharides, A. Lanitis, C. Pattichis, and M. Vento, Eds. Springer, Cham, Oct. 2021, pp. 275–284. https://doi.org/10.1007/978-3-030-89131-2_25

[11] R. Rothe, R. Timofte, and L. V. Gool, "Deep expectation of real and apparent age from a single image without facial landmarks," *International Journal of Computer Vision*, vol. 126, pp. 144–157, Apr. 2018. https://doi.org/10.1007/s11263-016-0940-3

[12] G. Guo, Y. Fu, C. R. Dyer, and T. S. Huang, "Image-based human age estimation by manifold learning and locally adjusted robust regression," *IEEE Transactions on Image Processing*, vol. 17, no. 7, pp. 1178–1188, July 2008. https://doi.org/10.1109/TIP.2008.924280

[13] P. O. Odiona, M. N. Musab, and S. U. Shuaibua, "Age prediction from sclera images using deep learning.," *Journal of the Nigerian Society of Physical Sciences*, vol. 4, no. 3, Aug. 2022. https://doi.org/10.46481/jnsps.2022.787

[14] X. Wang, R. Guo, and C. Kambhamettu, "Deeply-learned feature for age estimation," in *Proceedings of the 2015 IEEE Winter Conference on Applications of Computer Vision*, Waikoloa, HI, USA, Jan. 2015, pp. 534–541. https://doi.org/10.1109/WACV.2015.77

[15] Z. Niu, M. Zhou, L. Wang, X. Gao, and G. Hua, "Ordinal regression with multiple output CNN for age estimation," in *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Las Vegas, NV, USA, Jun. 2016, pp. 4920–4928. https://doi.org/10.1109/CVPR.2016.532

[16] S. Haseena, S. Saroja, R. Madavan, A. Karthick, B. Pant, and M. Kifetew, "Prediction of the age and gender based on human face images based on deep learning algorithm," *Computational and Mathematical Methods in Medicine*, vol. 2022, pp. 1–16, Aug. 2022. https://doi.org/10.1155/2022/1413597

[17] V. Sheoran, S. Joshi, and T. R. Bhayani, "Age and gender prediction using deep CNNs and transfer learning," in *Communications in Computer and Information Science*, S.K. Singh, P. Roy, B. Raman, and P. Nagabhushan, Eds. Springer Singapore, Mar. 2021, pp. 293–304. https://doi.org/10.1007/978-981-16-1092-9_25

[18] B. Agrawal and M. Dixit, "Age estimation and gender prediction using convolutional neural network," in *International Conference on Sustainable and Innovative Solutions for Current Challenges in Engineering & Technology*, 2019, pp. 163–175. https://doi.org/10.1007/978-3-030-44758-8_15

[19] A. Rattani, N. Reddy, and R. Derakhshani, "Convolutional neural networks for gender prediction from smartphone-based ocular images," *IET Biometrics*, vol. 7, pp. 423–430, Feb. 2018. https://doi.org/10.1049/iet-bmt.2017.0171

[20] U. Kulkarni, S. M. Meena, S. V. Gurlahosur, and G. Bhogar, "Quantization friendly MobileNet (QF-MobileNet) architecture for vision based applications on embedded platforms," *Neural Networks*, vol. 136, pp. 28–39, Apr. 2021. https://doi.org/10.1016/j.neunet.2020.12.022

[21] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015. MICCAI 2015. Lecture Notes in Computer Science*, vol 9351, N. Navab, J. Hornegger, W. Wells, and A. Frangi, Eds. Springer, Cham, 2015, pp. 234–241. https://doi.org/10.1007/978-3-319-24574-4_28

[22] M. Tan and Q. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proceedings of the 36th International Conference on Machine Learning*, Long Beach, California, USA, Jun. 2019, pp. 6105–6114. [Online]. Available: https://arxiv.org/pdf/1905.11946

[23] E. Korot, N. Pontikos, X. Liu, S. K. Wagner, L. Faes, J. Huemer, K. Balaskas, A. K. Denniston, A. Khawaja, and P. A. Keane, "Predicting sex from retinal fundus photographs using automated deep learning," *Scientific Reports*, vol. 11, May 2021, Art. no. 10286. https://doi.org/10.1038/s41598-021-89743-x

[24] S. W. P. Listio, "Performance of deep learning inception model and MobileNet model on gender prediction through eye image," *Sinkron*, vol. 6, no. 4, pp. 2593–2601, Oct. 2022. https://doi.org/10.33395/sinkron.v7i4.11887

[25] J. Hu, H.-J. Zeng, H. Li, C. Niu, and Z. Chen, "Demographic prediction based on user's browsing behavior," in *Proceedings of the 16th International Conference on World Wide Web*, *WWW'07*, May 2007, pp. 151–160. https://doi.org/10.1145/1242572.1242594

Hai Thanh Nguyen is a Lecturer of Information and Communication Technology College, Can Tho University, Vietnam. He received PhD degree in Computer Science from Sorbonne University, France. His current research includes bioinformatics, healthcare systems, recommendation systems, and applications using computer vision, machine learning, and data analysis.
E-mail: nthai.cit@ctu.edu.vn
ORCID iD: https://orcid.org/0000-0002-1386-1390

Linh Thuy Thi Pham is a Lecturer from the Faculty of Information Technology, Can Tho University of Technology. Her research interests are computational intelligence, artificial intelligence, biomedical engineering, image processing and computer vision.
E-mail: pttlinh@ctuet.edu.vn
ORCID iD: https://orcid.org/0009-0005-7338-0520

Dung Thi Dang is a Lecturer from the Faculty of Information Technology, Can Tho University of Technology. She began her postgraduate studies at the Faculty of Information Technology, Danang University of Technology, and graduated with a Master's degree in 2016. Her research interests include image processing and deep learning.
E-mail: dtdung@ctuet.edu.vn
ORCID iD: https://orcid.org/0009-0004-1874-4873

Quoc Thien Huynh Nguyen is a student in Computer Science, Can Tho University of Technology. His research interests include deep learning and image processing etc.
E-mail: nhtquoc2100802@student.ctuet.edu.vn
ORCID iD: https://orcid.org/0009-0003-3881-4376

Son Nguyen Huynh is a student in Computer Science, with research interests in artificial intelligence (AI), natural language processing (NLP), deep learning models, and software engineering. He is also a mobile application developer.
E-mail: huynhnguyenson2002@gmail.com
ORCID iD: https://orcid.org/0009-0008-5826-6318

Phu Hao Dang is a student in Computer Science at Can Tho University of Technology. He has experience in algorithms, image processing, and .NET.
E-mail: dhphu2100116@student.ctuet.edu.vn
ORCID iD: https://orcid.org/0009-0003-6592-6402